

WINTON, Australia

WMO: 943420

Lat: 22.3617S

Lon: 143.0836E

Elev: 634

StdP: 14.36

Time Zone: 10.00 (E10)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	42.8	45.3	12.3	10.5	66.7	15.9	12.6	67.9	22.0	66.7	20.7	65.5	5.4	90	0.430

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	23.6	105.8	69.3	103.7	69.2	101.6	69.3	77.8	88.9	76.8	88.0	75.9	87.6	9.2	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.6	137.0	80.6	74.4	131.8	80.5	73.1	125.9	80.7	41.8	88.3	40.8	88.0	39.9	87.2	90.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.6	19.9	17.8	DB	37.6	109.8	1.9	2.8	36.2	111.9	35.1	113.5	34.1	115.1	32.7	117.2	(4)
(5)			WB	31.5	81.0	2.3	2.8	29.8	83.1	28.5	84.7	27.2	86.3	25.5	88.4	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	76.8	87.3	85.8	83.2	77.2	69.5	63.5	63.1	65.8	73.9	80.5	84.5	87.3	(6)	
	DBStd	10.38	4.80	4.68	4.22	4.56	5.41	5.83	5.44	5.55	5.89	5.68	5.00	5.07	(7)	
	HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)	
	HDD65	286	0	0	0	1	21	98	103	57	5	1	0	0	(9)	
	CDD50	9765	1155	1003	1031	817	604	406	405	488	716	947	1036	1157	(10)	
	CDD65	4576	690	583	566	368	160	54	43	81	271	482	586	692	(11)	
	CDH74	60787	9896	8021	7217	4010	1514	485	515	1145	3424	6395	8123	10042	(12)	
(13)	CDH80	34289	6069	4741	3993	1911	456	81	88	353	1673	3669	4940	6315	(13)	
(14)	Wind	WSAvg	8.3	8.6	8.6	8.2	7.9	7.7	7.9	7.9	8.3	9.0	8.9	8.9	(14)	
(15) Precipitation	PrecAvg	15.6	2.9	3.7	1.9	0.8	0.8	0.5	0.5	0.3	0.3	0.7	1.3	2.3	(15)	
	PrecMax	46.1	22.2	18.3	9.6	7.1	3.9	3.7	4.4	1.9	1.4	4.8	9.6	7.8	(16)	
	PrecMin	2.1	0.0	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)	
	PrecStd	7.6	3.7	3.9	2.3	1.5	1.1	0.9	1.0	0.4	0.4	1.0	2.1	1.8	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	108.2	106.8	103.2	97.1	91.3	85.8	86.2	92.2	99.7	104.7	106.5	108.7	(19)	
		MCWB	70.5	70.4	70.0	68.2	65.4	64.4	63.5	63.1	65.1	66.0	68.1	69.5	(20)	
	2%	DB	104.8	103.0	100.3	94.2	87.4	82.4	82.1	87.4	95.8	101.1	103.6	105.8	(21)	
		MCWB	70.7	71.6	70.1	67.2	64.5	62.6	60.8	61.3	63.6	66.1	67.7	68.9	(22)	
	5%	DB	102.1	100.2	97.6	91.9	84.9	79.8	79.5	83.9	92.6	98.3	101.2	103.4	(23)	
		MCWB	71.1	71.7	69.9	66.5	63.3	61.6	59.6	59.4	62.6	65.2	67.0	68.9	(24)	
	10%	DB	99.5	97.8	95.0	89.7	82.2	76.6	76.6	80.6	89.5	95.3	98.5	100.7	(25)	
		MCWB	71.2	71.6	69.6	65.9	62.5	59.5	57.9	58.2	61.5	64.5	66.9	69.4	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	79.3	78.7	78.1	75.1	70.3	67.4	68.7	67.1	71.2	73.1	76.2	78.4	(27)	
		MCDB	90.8	89.1	85.7	86.0	81.5	80.2	74.4	83.0	83.2	88.6	90.1	93.2	(28)	
	2%	WB	77.8	77.4	76.5	73.1	68.2	65.2	65.2	64.5	68.5	70.8	74.1	76.3	(29)	
		MCDB	89.1	88.7	85.1	85.2	80.8	77.2	77.7	80.0	85.4	87.0	87.9	89.3	(30)	
	5%	WB	76.8	76.5	75.3	71.2	66.2	63.5	62.1	62.1	66.4	69.2	72.7	75.1	(31)	
		MCDB	88.0	88.4	85.4	84.2	80.0	75.3	74.8	78.3	84.3	87.2	86.9	88.7	(32)	
	10%	WB	75.7	75.6	74.0	69.2	64.4	61.6	59.7	60.0	64.1	67.7	71.3	74.1	(33)	
		MCDB	87.4	88.0	85.9	83.0	78.4	73.2	74.0	77.4	83.7	87.5	86.5	88.2	(34)	
(35) Mean Daily Temperature Range	5% DB	MDBR	21.8	21.9	22.7	24.2	24.7	24.3	26.5	28.6	29.0	27.4	25.5	23.6	(35)	
		MCDBR	25.6	25.3	25.8	26.4	26.4	26.5	29.4	32.2	32.1	30.5	28.7	27.5	(36)	
	5% WB	MCWBR	8.5	7.7	8.7	11.0	12.2	13.0	15.0	15.9	14.4	12.7	11.3	10.5	(37)	
		MCWBR	18.8	19.5	18.7	21.2	22.3	22.6	23.8	26.8	25.6	24.9	22.6	21.2	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.386	0.375	0.357	0.345	0.331	0.310	0.307	0.319	0.362	0.378	0.387	0.399	(40)	
		taud	2.466	2.495	2.526	2.509	2.489	2.539	2.525	2.469	2.366	2.346	2.369	2.378	(41)	
	Ebn at Noon		304	303	300	291	282	283	288	295	294	298	302	300	(42)	
		Edh at Noon	38	36	34	32	30	27	28	32	39	42	42	42	(43)	
(44) All-Sky Solar Radiation	RadAvg		2272	2246	2061	1805	1485	1323	1462	1768	2068	2327	2389	2350	(44)	
	RadStd		189	127	151	148	76	91	77	73	149	142	126	148	(45)	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
(47)	Regional (2 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page